

Nirmalya Ghosh

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EXPERIENCE

Teradata

Senior Cloud Services Engineer

Seattle, WA - Remote

Mar 2026 - Present

- **Architect and lead delivery** of core cloud control-plane services across **AWS, Azure, and GCP**, driving elastic scalability and on-demand provisioning, and disaster-recovery capabilities for Teradata's managed cloud offerings.
- **Spearhead cross-cloud Private Service Connect (PSC) automation**, designing subnet allocation strategies and forwarding-rule lifecycle management that reduced manual provisioning effort by over **60%**.
- **Lead fleet-wide OS configuration and certificate distribution** across GCP VM fleets using OS Config Agent and Policy Orchestration, ensuring consistent TLS trust anchors across hundreds of nodes with zero manual intervention.
- **Design and implement multi-interface networking** on SUSE Linux (SLES) VMs, resolving complex Cloud NAT, route-table, and DNS conflicts in multi-tenant architectures spanning multiple cloud regions.
- **Define technical direction and mentor** junior and mid-level engineers on Terraform IaC patterns, GCP networking, and secure-by-default infrastructure design principles, improving team velocity and code quality.
- **Drive cross-team collaboration** with security, platform, and application engineering teams to align infrastructure changes with compliance and operational requirements across multi-cloud environments.

Teradata

Cloud Services Engineer

Seattle, WA - Remote

Jan 2024 - Mar 2026

- Developed and owned the **Kubernetes-level, multi-cloud Network Service**, delivering essential networking capabilities like **IP whitelisting**, GCP subnet creation, **Express Route**, and **VPN connectivity**.
- Built a plugin-based **credential rotation service** using the Template Method design pattern in Python, integrating with **HashiCorp Vault** to automate rotation for Valtix API keys and GCP service account keys — reducing credential downtime incidents by **75%** and strengthening security compliance.
- Designed and implemented network-related **Terraform** configurations to support and extend Network Service features, ensuring seamless integration and consistent infrastructure automation across multi-cloud platforms.
- Executed **Terraform state migration** from GitLab to AWS S3 backends across staging, pre-prod, and production environments, establishing durable remote state with DynamoDB locking.
- Configured and maintained **CI/CD pipelines** using **GitHub Actions** and **JFrog Artifactory**, enabling reliable artifact promotion and reproducible infrastructure deployments.

Teradata

Cloud Software Engineer Intern

Seattle, WA - Remote

June 2023 - December 2023

- Built a **Network Performance Evaluation Microservice** for SREs to detect and diagnose network issues, cutting resolution time by **40%** and improving operational efficiency by **30%**
- Developed **AWS Lambda functions** in Python with **Terraform** to automate VM provisioning across AWS and Azure, reducing manual provisioning time by **50%** and human error by **40%**
- Extended a **FastAPI** microservice to support AWS and Azure Capacity Reservation Groups (reducing infrastructure costs by **15%**) and expanded product availability to Azure *westus* and *westeurope*, growing the user base by **25%**

Teradata

Cloud Software Engineer Intern

Seattle, WA - Remote

June 2022 - September 2022

- Built a **FastAPI** service using **Boto3** to query AWS Service Quotas and compute utilization metrics, serving as a pre-check gate for customer cloud provisioning
- Developed automated **end-to-end API tests** with **Postman** integrated into **GitLab CI/CD**, validating functionality, performance, and reliability

PERSONAL PROJECTS

LetsGoSEA Web Application | 🌐 Azure | 🌐

ASP.NET, C#, Razor, Javascript, Bootstrap CSS, NUnit

- Created a **full-stack** web application that allows users to explore beautiful Seattle neighborhoods
- Implemented **RESTful APIs** with **CRUD** features including rate, comment, and upload/delete images.
- Integrated **Google Map API** to display neighborhood location. Conducted defensive programming in test-driven development and achieved **100% code coverage**

Maze Runner | 🌐

Python, Socket, Multithreading

- **Led** development of a multiplayer maze runner game for users to compete against each other to finish a random maze
- Managed authentication, query handling, and data transfer with **Socket**, **TCP/IP**, and **RPCs**. Implemented multiplayer concurrency using **thread pool** and **mutex lock**.

EDUCATION

Master of Science: Computer Science (Software Engineering)

December 2023

Seattle University

Seattle, WA

- Award: Winner of ACM Club Project Competition, Winter '21, Werewolf Online Game

Bachelor of Arts: Geography

Sept 2016 - Mar 2020

University of Washington

Seattle, WA

- Award: Senior Year Dean's List
- Leadership: Community Manager - Plenum (Undergraduate Geography Research Journal)

SKILLS

- Python, FastAPI, REST API, Docker, Kubernetes, Terraform, HashiCorp Vault, Cloud Architecture & Orchestration, AWS (Lambda, S3, EC2, APIGateway, DynamoDB, Route53), Azure (VNet, Express Route, AKS), GCP (GKE, Cloud NAT, PSC), GitHub Actions, JFrog Artifactory, Shell Scripting, Javascript, ReactJS, NodeJS, C++, PostgreSQL, MySQL, Git, Linux